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United States Patent	12402282
Kind Code	B2
Date of Patent	August 26, 2025
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Liquid cooling unit

Abstract

A liquid cooling unit includes a first flow divider/combiner, a second flow divider/combiner, a plate heat exchanger, a radiator and a heater. The plate heat exchanger, the radiator and the heater are connected in parallel between a flow dividing end of the flow divider/combiner and a flow combining end of the second flow divider/combiner. The flow dividing end of the first flow divider/combiner outputs a fluid into the plate heat exchanger, the radiator or the heater. The plate heat exchanger and the radiator cool the fluid, and the heater heats the fluid. The flow combining end of the second flow divider/combiner receives the fluid outputted by the plate heat exchanger, the radiator or the heater, and the flow dividing end of the second flow divider/combiner outputs the fluid to a battery for thermal management of the battery. The flow combining end of the first flow divider/combiner receives the fluid.

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Appl. No.: 18/343834

Filed: June 29, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230345674 A1	Oct. 26, 2023

Foreign Application Priority Data

CN	202122794342.3	Nov. 15, 2021
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Related U.S. Application Data

continuation parent-doc WO PCT/CN2022/097112 20220606 PENDING child-doc US 18343834

Publication Classification

Int. Cl.: H01M10/6551 (20140101); H01M10/615 (20140101); H01M10/66 (20140101); H05K7/20 (20060101)

U.S. Cl.:

CPC H05K7/20381 (20130101); H05K7/20327 (20130101);

Field of Classification Search

CPC: H01M (10/06)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of International Patent Application No. PCT/CN2022/097112, filed on Jun. 6, 2022, which claims the priority of Chinese Patent Application No. 202122794342.3, filed on Nov. 15, 2021. The disclosures of the aforementioned patent applications are incorporated herein in by reference in their entirety.

TECHNICAL FIELD

(1) The application relates to the technical field of liquid cooling device, and more particularly relates to a liquid cooling unit.

BACKGROUND

(2) To improve the power supply performance of batteries, a plurality of batteries are usually connected in series or in parallel to form a battery pack for use. However, the plurality of batteries electrified simultaneously will cause a heating phenomenon, and particularly in a high-temperature environment, the heating phenomenon is more severe, which may result in combustion and even explosion thanks to overheat of the batteries. Thus, there is a severe potential safety hazard.

(3) Water cooling uniformly and stably takes away heat of the batteries through a water flow, which is a frequently used cooling mode. However, existing water cooling structures often occupy large space and are poor in cooling efficiency. Therefore, it is an urgent need of a water cooling device with high cooling efficiency.

SUMMARY

(4) Embodiments of the application provide a liquid cooling unit which is capable of providing various cooling modes and connecting different cooling devices in parallel, thereby improving the cooling efficiency of the liquid cooling unit.

(5) The embodiments of the application provide a liquid cooling unit, including a first flow divider/combiner, a second flow divider/combiner, a plate heat exchanger, a radiator and a heater, where the plate heat exchanger, the radiator and the heater are connected in parallel between a flow dividing end of the flow divider/combiner and a flow combining end of the second flow divider/combiner; the flow dividing end of the first flow divider/combiner is configured to output a fluid into at least one of the plate heat exchanger, the radiator and the heater; the plate heat exchanger and the radiator are configured to cool the fluid, and the heater is configured to heat the fluid; the flow combining end of the second flow divider/combiner is configured to receive the fluid outputted by at least one of the plate heat exchanger, the radiator and the heater, and the flow dividing end of the second flow divider/combiner is configured to output the fluid to a battery for thermal management of the battery; and the flow combining end of the first flow divider/combiner is configured to receive the fluid after thermal management of the battery.

(6) It is to be noted that the above thermal management refers to cooling or heating the battery through the fluid.

(7) In some embodiments, the liquid cooling unit further includes a compressor, a condenser and a cooling fan, where the compressor, the condenser, the plate heat exchanger and the cooling fan are configured to cool a refrigerant; and where a refrigerant pipeline connected to the compressor, the condenser and the plate heat exchanger are integrally welded.

(8) In the above implementation modes, the compressor, the condenser, the radiator and the plate heat exchanger are connected by utilizing integrally formed refrigerant pipelines by welding to form an integrated compressor refrigeration model, so that the numbers of the refrigerant pipelines and other parts assembled are reduced, and the installation efficiency of the liquid cooling unit is improved.

(9) In some embodiments, the liquid cooling unit further includes a fluid chamber, where the fluid

chamber is a closed fluid chamber and is configured to provide a fluid to the second flow divider/combiner.

(10) It should be understood that the fluid chamber provided by the application can also be known as a water tank, the name of which is not defined herein.

(11) In the above implementation modes, the fluid chamber provides a cooling fluid to the pipeline through the second flow divider/combiner. By adopting the closed fluid chamber, it can be ensured that a cooling fluid system is sealed during operation, which is beneficial to guaranteeing cleanness, stability and long-term action of the cooling fluid.

(12) In some embodiments, the liquid cooling unit further includes a water pump, where the water pump is configured to output the fluid to the battery.

(13) In some embodiments, the water pump is a brushless direct current water pump. Without carbon brush friction, the brushless direct current water pump does not generate sparks and features high efficiency, low power consumption, long service life compared with the brush motor and low noise.

(14) In some embodiments, there are a plurality of heaters.

(15) In the above implementation modes, the number of the heaters can be set according to an actual demand. For example, in a case where the battery is in an environment with relatively low environment temperature, the plurality of heaters can be arranged and are connected in parallel between the first flow divider/combiner and the second flow divider/combiner to jointly heat the battery, so as to improve the heating efficiency of the liquid cooling unit.

(16) In some embodiments, the plate heat exchanger, the radiator and the heaters are connected to the first flow divider/combiner and the second flow divider/combiner through rubber hoses.

(17) It is to be noted that the pipelines connecting the plate heat exchanger, the heat radiator, the heaters, the fluid chamber, the water pump, the first flow divider/combiner and the second flow divider/combiner in the liquid cooling unit all are rubber hoses to form a cooling fluid loop of the liquid cooling unit.

(18) In the above implementation modes, corresponding numbers of heaters and water pumps can be connected according to the actual demand of thermal management of the battery, so that the installation flexibility of parts in the liquid cooling unit is improved.

(19) In some embodiments, the first flow divider/combiner and the second flow divider/combiner both are provided with normally closed drain valves, configured to conduct drainage and exhaust of a connecting pipeline connected to the first flow divider/combiner and the second flow divider/combiner.

(20) In some embodiments, the first flow divider/combiner and the second flow divider/combiner both are integrally formed by welding.

(21) In some embodiments, in a case where an environment temperature is lower than a first threshold, the cooling fan is started to cool the fluid inside the radiator.

(22) It should be understood that to start the cooling fan is to cool the cooling fluid inside the radiator, so as to cool the battery. In default, the water pump cools the cooling fluid only in the started state. The power consumption of the cooling fan and the water pump working jointly is far less than that of the compressor refrigeration module. Power of the cooling fan and the water pump is usually hectowatt, and the electric power of the compressor refrigeration module is higher than kilowatt. Therefore, to start the cooling fan and the water pump for cooling may reduce the power consumption of the liquid cooling unit.

(23) In the above implementation mode, in a case where the external air temperature is lower than the temperature of the cooling fluid in the cooling fluid loop, for example, in a case where the environment temperature is lower than a certain threshold, the radiator and the water pump can be started forcibly. The cooling fan ventilates to cool the cooling fluid so as to assist the system for cooling, so that the power consumption of the unit is reduced.

(24) In the embodiments of the application, by designing the parallel structure among the plate heat

exchanger, the radiator and the heaters, the liquid cooling unit can select different devices to work according to the demand of the battery on thermal management. In some embodiments, the battery can be jointly cooled through branches of the plate heat exchanger and the radiator, so that the cooling efficiency of the liquid cooling unit is improved.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) To describe the technical solutions of the embodiments of the application clearer, drawings needed to be used in the embodiments of the application will be briefly described below. Apparently, the drawings described below are only some embodiments of the application. Those of ordinary skill in the art can further obtain other drawings according to the drawings without making creative efforts.
- (2) FIG. 1 is a block diagram of a liquid cooling unit provided in some embodiments of the application.
- (3) FIG. 2 is a connecting schematic diagram of a cooling fluid loop of the liquid cooling unit provided in some embodiments of the application.
- (4) FIG. 3 is a structural schematic diagram of the liquid cooling unit provided in some embodiments of the application.

DETAIL DESCRIPTION

- (5) The technical solutions in the embodiments of the application are described below in combination with drawings.
- (6) The embodiments of the technical solutions of the application will be described in detail below in combination with drawings. The embodiments below are merely used to more clearly describe the technical solutions of the application, merely as examples, instead of limiting the scope of protection of the application.
- (7) Unless otherwise defined, all technical and scientific terms used herein are identical to meaning commonly understood by those skilled in the art of the application. The terms used herein are merely used to describe specific embodiments rather than limiting the application. The terms “include” and “have” and any variation thereof in the description, claims and the above description of drawings of the application are intended to cover non-exclusive inclusion.
- (8) In description of the application, terms “first”, “second” and the like are merely used to distinguish different objects rather than being construed to indicate or imply relative importance or implicitly indicate the quantity, specific order or primary and secondary relation of indicated technical features. In the description of the application, “a plurality of” means two or more, unless expressly specified otherwise.
- (9) The “embodiments” in the application mean that specific features, structure or characteristics described in combination with the embodiments may be included in at least one embodiment of the application. The phrase emerges in each position of the description is not necessarily the same embodiment or independent or alternative embodiments mutually exclusive to other embodiments. Persons skilled in the art explicitly and implicitly understand that the embodiments described herein may combine with other embodiments.
- (10) In the description of the application, the term “and/or” is only an association relationship describing associated objects and represents that three relationships may exist. For example, A and/or B may represent three conditions: i.e., independent existence of A, existence of both A and B and independent existence of B. In addition, character “/” in the disclosure usually represents that previous and next associated objects form an “or” relationship.
- (11) In the description of the application, term “a plurality of” means more than two (including two), and similarly, “a plurality of groups” means more than two groups (including two groups),

and “a plurality of sheets” means more than two sheets (including two sheets).

(12) In the description of the application, orientation or position relations indicated by the technical terms “central”, “longitudinal”, “transverse”, “length”, “width”, “thickness”, “upper”, “lower”, “front”, “back”, “left”, “right”, “vertical”, “horizontal”, “top”, “bottom”, “inner”, “outer”, “clockwise”, “anticlockwise”, “axial”, “radial”, “circumferential”, and the like are orientation or position relations based on the drawings only for ease of description of the application and for simplicity of description, and are not intended to indicate or imply that the referenced device or element must have a particular orientation and be constructed and operative in a particular orientation, and thus may not be construed as a limitation on the present application.

(13) In the description of the application, unless otherwise specified and defined, the technical terms such as “mount”, “connect”, “connection” and “fix” shall be understood in a broad sense. For example, it can be either fixed connection or detachable connection or integrated connection; either mechanical connection or electrical connection with each other; either direct connection or indirect connection through an intermediate, and communication in two components or interaction relaxation of the two components.

(14) Those of ordinary skill in the art can understand specific meaning of the terms in the disclosure under specific circumstances. The batteries are usually used as a plurality of batteries are connected in series or in parallel to form a battery pack, so as to enhance the power supply performance of the batteries. Box batteries have ultra-high capacities. However, the plurality of batteries electrified simultaneously will cause a heating phenomenon, the various heating phenomena of the battery pack is more severe, which may result in combustion and even explosion thanks to overheat of the batteries. Thus, there is a severe potential safety hazard, which affects the service lives of the batteries. Current cooling modes for batteries are usually divided into two types. An air cooling mode mainly relies on external cold air to take away heat on the surfaces of the batteries to cool the batteries. The process is affected by many uncertain factors such as air speed of an air flow and is likely to cause non-uniform radiation, thereby resulting in low refrigerating and heating efficiencies. A water cooling mode cools the batteries as a water flow takes away heat of the batteries, and this mode features high stability. However, current water cooling structures or systems often have the problem of large occupied area and low cooling efficiency, and with respect to part type liquid cooling unit, parts thereof need to be installed and connected on site in a power station, so that the field installation and maintenance difficulty and cost are improved.

(15) Aiming at the above problems, the embodiments of the application provide a liquid cooling unit. All parts are integrated in an integrated unit to solve the problem of cold leakage caused by large occupied space of the current water cooling structures, so that the radiating efficiency of the battery pack is improved. In addition, the integrated liquid cooling unit is free of field installation, so that the field installation and maintenance difficulty and cost are lowered. The liquid cooling unit provided by the application is suitable for thermal management of various batteries or battery packs, so that the temperature of the batteries or battery packs can be controlled in a reasonable range.

(16) FIG. 1 is a structural diagram of a liquid cooling unit **100** provided by the embodiments of the application for thermal management of the battery **200**. The liquid cooling unit **100** includes a first flow divider/combiner **11**, a second flow divider/combiner **12**, a plate heat exchanger **13**, a radiator **14** and a heater **15**, where the plate heat exchanger **13**, the radiator **14** and the heater **15** are connected in parallel between a flow dividing end of the flow divider/combiner **11** and a flow combining end of the second flow divider/combiner **12**; the flow dividing end of the first flow divider/combiner **11** is configured to output a fluid into at least one of the plate heat exchanger **13**, the radiator **14** and the heater **15**; the plate heat exchanger **13** and the radiator **14** are configured to cool the fluid, and the heater **15** is configured to heat the fluid; the flow combining end of the second flow divider/combiner **12** is configured to receive the fluid outputted by at least one of the plate heat exchanger **13**, the radiator **14** and the heater **15**, and the flow dividing end of the second

flow divider/combiner **12** is configured to output the fluid to a battery **200** for thermal management of the battery **200**; and the flow combining end of the first flow divider/combiner **11** is configured to receive the fluid after thermal management of the battery **200**.

(17) The first flow divider/combiner **11** and the second flow divider/combiner **12** can be also known as flow divider/combiner valves. The flow divider valves play a role of supplying a same flow (equally divided) to more than two execution elements from a same oil source in a hydraulic system or supplying a flow (proportionally divided) to the two execution elements in a certain proportion, so as to keep synchronization of speeds of the two execution elements or keep a definite proportion relation. The flow combiner valves play a role of collecting equal flow or proportional oil return flow from the two execution elements to achieve speed synchronization or definite proportion relation therebetween. The flow divider/combiner valves have the functions of the flow divider valves and the flow combiner valves.

(18) The plate heat exchanger **13** is a heat exchanger formed by the following steps: pressing thin metal plates to heat exchange plates with a certain ripple form; then overlapping the heat exchange plates; and fastening the heat exchange plates with clamping plates and bolts. Thin rectangular channels are formed among various plates for heat exchange through semi-plates. A working fluid flows in a narrow and zigzag channel formed between two plates. Cold and hot fluids pass through the channel in sequence, and the fluids are separated by a middle spacer plate and exchange heat through the plate.

(19) The radiator **14** is of a parallel fluid finned tube structure made from an aluminum material, the fluids circulate inside a row flow, a parallel channel thereof is of a harmonica-shaped tube structure, and thin fins are brazed to the surface, so that the radiating area can be increased.

(20) It should be understood that thermal management in the embodiments of the application refers to cooling or heating the battery **200** through the fluids.

(21) Through the implementation mode, the plate heat exchanger **13**, the radiator **14** and the heaters **15** inside the liquid cooling unit **100** are of a parallel structure, and different branches can be selected according to different demands of the battery **200** on heating or refrigerating to work. In some embodiments, the battery can further be jointly cooled by branches of the plate heat exchanger **13** and the radiator **14**, so that the cooling efficiency of the liquid cooling unit **100** is improved.

(22) In some embodiments, the liquid cooling unit **100** further includes a compressor **16**, a compressor **17** and a cooling fan **18**, where the compressor **16**, the compressor **17**, the cooling fan **18** and the plate heat exchanger are configured to cool the refrigerant. A refrigerant pipeline **19** connected to the compressor **16**, the compressor **17** and the plate heat exchanger **13** is integrally welded.

(23) The compressor **17** is a driven fluid machine which boots a low pressure gas to a high pressure gas, which is the heart of the refrigeration system. The compressor sucks the low-temperature and low-pressure refrigerant gas from a gas suction pipe, and the motor runs to drive a piston to compress the refrigerant gas. The compressor discharges a high-temperature and high-pressure refrigerant liquid to an exhaust pipe so as to provide power to refrigeration cycle. The compressor **17** can convert gas or steam into liquid, and transfers heat in the pipe to air surrounding the pipe rapidly.

(24) It is to be noted that the compressor **16**, the compressor **17**, the cooling fan **18**, the plate heat exchanger **13** and the refrigerant pipeline **19** form the compressor refrigeration module **110** of the liquid cooling unit **100** in the embodiments of the application.

(25) In the above implementation modes, the refrigeration pipeline is integrally formed by welding, so that the numbers of a plurality of parts assembled in the refrigerant pipeline **20** and the compressor refrigeration module **110** are reduced, and the installation and maintenance efficiencies of the liquid cooling unit **100** are improved.

(26) It is to be further noted that the liquid circulating in the refrigerant pipeline **19** in the

compressor refrigeration module **110** of the liquid cooling unit **100** is the refrigerant. Besides pipelines inside the compressor refrigeration module **110** in the liquid cooling unit **100**, the liquids circulating in other pipelines are cooling fluids.

(27) It is to be further noted that the radiator **14**, the cooling fan **18** and a cooling fluid pipeline matched therewith form the radiator refrigeration module **120** of the liquid cooling unit **100**. The first flow divider/combiner **11**, the second flow divider/combiner **12** and the heaters **15** form the heating and cooling fluid pipeline module **130** of the liquid cooling unit **10** in the embodiments of the application.

(28) In some embodiments, the liquid cooling unit further includes a fluid chamber **20**, where the fluid chamber is a closed fluid chamber and is configured to provide a fluid to the second flow divider/combiner **12**.

(29) It should be understood that the fluid chamber **20** can also be known as a water tank, which is configured to store the fluids.

(30) In the embodiments of the application, the fluids refer to the cooling fluids.

(31) In the above implementation modes, the fluid chamber **20** can be a closed fluid chamber, and it can be ensured that a cooling fluid system is sealed during operation, which is beneficial to guaranteeing cleanness, stability and long-term action of the cooling fluid.

(32) In some embodiments, the fluid chamber **20** can be a closed expanded fluid chamber which can either buffer pressure fluctuation in the system or play an unloading role. In a case where the water pressure in the system changes, the change of the water pressure will be induced thanks to its automatic expanding and shrinking function, thereby playing a buffering role. The expanded fluid chamber can maintain the water pressure stable so as to control the water to not change due to pressure.

(33) In some embodiments, the liquid cooling unit **100** further includes a water pump **21**, where the water pump is configured to output the fluid to the battery **200** so as to cool or heat the battery **200**.

(34) In some embodiments, there are a plurality of water pumps **21** in the liquid cooling unit **100**. For example, the liquid cooling unit **100** shown in FIG. 1 includes five water pumps. The five water pumps are connected in parallel to the second flow divider/combiner **12** and each of the water pumps is connected to one battery **200**, so as to cool or heat the battery **200** connected thereto.

(35) In some embodiments, the water pumps **21** are brushless direct current water pumps. Without carbon brush friction, the brushless direct current water pumps do not generate sparks and feature high efficiency, low power consumption, long service life compared with the brush motor and low noise.

(36) It is to be noted that the water pumps **21** are a power source of the whole cooling fluid circulation system, which can control the outlet flow to ensure that peripheral cooling objects (batteries, battery packs, battery boxes and the like) have enough cooling fluid flows and to ensure the consistency of fed flows.

(37) In some embodiments, the water pumps **21** are pulse width modulation (PMW) variable frequency regulation water pumps, which are beneficial to regulating and outputting different cooling fluid flows according to different working condition requirements of the peripheral pipeline system on cooling fluid objects.

(38) In the embodiments of the application, the number of water pumps **21** can be selectively configured according to the number of peripheral batteries **200** to facilitate consideration of demands of various client systems. For example, when a client needs to cool or heat four batteries **200**, the number of water pumps **21** in the liquid cooling unit **100** can be set as 4. Of course, the number of water pumps **21** can be subject to redundancy design as well. For example, when a client needs to cool or heat four batteries **200**, the number of water pumps in the liquid cooling unit **100** can be set as 5; at the time, the fifth water pump is closed only, and the waterway is plugged with a plug pipeline. In a case where a certain water way fails later, it is adapted to the fifth water pump to

recover use rapidly without replacing the unit with a new unit.

(39) In some embodiments, the plate heat exchanger **13**, the radiator **14** and the heaters **15** are connected to the first flow divider/combiner **11** and the second flow divider/combiner **12** through rubber hoses.

(40) In some embodiments, the water pumps **21** are connected to the second flow divider/combiner **12** and the batteries **200** through the rubber hoses as well.

(41) In some embodiments, the first flow divider/combiner **11** and the second flow divider/combiner **12** both are provided with normally closed drain valves, configured to conduct drainage and exhaust of a connecting pipeline connected to the first flow divider/combiner and the second flow divider/combiner.

(42) During first time fluid injection or fluid supplementation in after-sales maintenance, in a case where there are bubbles in the pipelines inside the liquid cooling unit **100**, the flowing stability of the cooling fluid inside the fluid cooling unit **100** and the water pressure sampling stability will be affected. By arranging the normally closed drain valves on the first flow divider/combiner **11** and the second flow divider/combiner **12**, drainage and exhaust of the pipelines inside the liquid cooling unit **100** are conducted.

(43) In some embodiments, the first flow divider/combiner **11** and the second flow divider/combiner **12** both are integrally formed by welding.

(44) In some embodiments, in a case where an environment temperature is lower than a first threshold, the cooling fan **18** is started to cool the fluid inside the radiator **14**.

(45) It should be understood that in a case where the liquid cooling unit **100** works in an environment with the temperature lower than a certain threshold, for example 15° C., the cooling fluid can be cooled by utilizing the radiator refrigeration module **120**, so that the batteries **200** are cooled. The specific cooling way by utilizing the radiator refrigeration module **120** refers to starting the cooling fan **18** and the water pumps **21**, where the cooling fan **18** ventilates to cool the cooling fluid inside the row flow of the radiator. The radiator refrigeration module **120** can either work independently or jointly work with the refrigerant refrigeration module **130** to cool the batteries **200**, which is not defined herein.

(46) In the above implementation modes, the power consumption of the cooling fan **18** and the water pumps **21** working jointly is far less than that of the compressor refrigeration module **110**. Power of the radiator and the water pumps is usually hectowatt, and the electric power of the compressor refrigeration module **110** is higher than kilowatt. Therefore, the power consumption of the liquid cooling unit **100** can be reduced by utilizing the radiator refrigeration module **120**.

(47) To facilitate understanding of the flow direction of the cooling fluid of the liquid cooling unit in the embodiments of the application, FIG. 2 shows the cooling fluid loop of the liquid cooling unit **100** provided by the embodiments of the application. The connecting pipelines all in FIG. 2 are cooling fluid pipelines, and arrows are flow directions of the cooling fluid.

(48) In some embodiments, there are a plurality of heaters **15**.

(49) Through the implementation mode, the number of the heaters **15** can be set according to the actual demand. For example, when the batteries are in an environment with low temperature, the plurality of heaters **15** can be arranged and connected in parallel between the first flow divider/combiner **11** and the second flow divider/combiner **12** to heat the batteries **200** jointly, so that the heating efficiency of the liquid cooling unit **100** is improved.

(50) Exemplarily, there are two heaters **15** in the liquid cooling unit **100** shown in FIG. 2, and the heaters are connected in parallel between the first flow divider/combiner **11** and the second flow divider/combiner **12**. Therefore, the first flow divider/combiner **11** includes four flow dividing ends. Further, it can be known that when there are three heaters, the first flow divider/combiner **11** includes five flow dividing ends. The flow dividing ends of the first flow divider/combiner can be set or manufactured as required.

(51) To facilitate understanding of the working flow of the refrigeration and cooling fluid pipeline

in the liquid cooling unit **100**, the working flow is introduced by the following embodiments.

(52) In some embodiments, in a case where the temperature of the battery **200** is higher than a certain threshold, in order to prevent thermo-runaway of the battery **200**, the battery **200** needs to be cooled. At the time, the water pump **21** corresponding to the battery is opened, the fluid chamber **20** provides the cooling fluid to the second flow divider/combiner **12**, and the cooling fluid flows to the battery **200** from the water pump **21** so as to cool the battery **200**. The cooling fluid cooling the battery **200** flows into the first flow divider/combiner **11** through the cooling fluid pipeline **300** outside the liquid cooling unit **100**. At the time, whether the external temperature is less than the first threshold is determined. In a case where the external temperature is less than the first threshold, the first flow divider/combiner **11** outputs the cooling fluid to the plate heat exchanger **13** and the radiator **14**, and the cooling fan **18** is started. In the branch of the plate heat exchanger, the cooling fluid flowing through is cooled by the plate heat exchanger **13**; in the branch of the radiator, the cooling fan **18** cools the cooling fluid inside the row flow of the radiator **14** and then transports the cooled cooling fluid to the second flow divider/combiner **12** to further circularly cool the battery **200**. In a case where the external temperature is not less than the first threshold, the first flow divider/combiner **11** only outputs the cooling fluid to the plate heat exchanger **13**.

(53) In some embodiments, in a case where the temperature of the battery **200** is LOWER than a certain threshold, for example, the temperature of the battery **200** is lower than the temperature of the cooling fluid, the battery **200** needs to be heated. At the time, the water pump **21** corresponding to the battery is opened, the fluid chamber **20** provides the cooling fluid to the second flow divider/combiner **12**, and the cooling fluid flows to the battery **200** from the water pump **21** so as to heat the battery **200**. The cooling fluid heating the battery **200** flows into the first flow divider/combiner **11** through the cooling fluid pipeline **300** outside the liquid cooling unit **100**. The first flow divider/combiner **11** outputs the cooling fluid to the heaters **15**, the heaters **15** heat the cooling fluid, and then the heated cooling fluid is transported to the second flow divider/combiner **12** to further circularly heat the battery **200**.

(54) It is to be noted that when the battery **200** is heated, the first flow divider/combiner **11** can output the cooling fluid to one or more heaters **15** according to actual working condition requirements, which is not defined herein.

(55) FIG. 3 shows a structural schematic diagram of a liquid cooling unit **100** in the embodiments of the application.

(56) In some embodiments, the way of assembly of parts inside the liquid cooling unit **100** is shown in FIG. 3. All parts of the liquid cooling unit **100** are fixed into a metal plate frame **22** provided with a forklift port **23** for facilitating transportation or carrying the liquid cooling unit **100** to an application site.

(57) In some embodiments, the interior of the liquid cooling unit **100** can be divided into three layers: the radiator **14**, the compressor **17**, the fluid chamber **20** and the cooling fan **18** in the uppermost layer; the plate heat exchanger **13** and the compressor **16** in the middle layer; and the first flow divider/combiner **11**, the second flow divider/combiner **12**, the heaters **15** and the water pumps **21** in the bottommost layer.

(58) The fluid chamber **20** is arranged in the uppermost layer inside the liquid cooling unit **100**, which is beneficial to improving the water pressure.

(59) In other embodiments, fixed positions of the above parts all are merely exemplary, and the positions and ways in actual fixation can be re-arranged according to working conditions or client demands, which is not defined herein.

(60) In the above implementation modes, all parts cooling or heating the batteries **200** form the integrated liquid cooling unit **100**, which solves the problems of complex pipelines, low cooling efficiency and the like caused by a conventional split indoor-outdoor unit design solution, and further solves the problem that the part type liquid cooling unit features large occupied space and high field installation and maintenance costs. In some embodiments, it is beneficial to controlling

batch production of the liquid cooling unit **100**, so that the liquid cooling unit has industrial use value.

(61) The above embodiments are merely used to describe the technical scheme of the application and are not used to limit the application. For those skilled in the art, various alternations and changes can be made on the application. Any modification, equivalent replacement, improvement, etc. made within the spirit and principle of the application shall be regarded as within the scope of the application.

Claims

1. A liquid cooling unit, comprising: a first flow divider/combiner; a second flow divider/combiner; a plate heat exchanger; a radiator; and a heater; wherein the plate heat exchanger, the radiator and the heater are connected in parallel between a flow dividing end of the first flow divider/combiner and a flow combining end of the second flow divider/combiner; the flow dividing end of the first flow divider/combiner is configured to output a fluid into at least one of the plate heat exchanger, the radiator and the heater; the plate heat exchanger and the radiator are configured to cool the fluid, and the heater is configured to heat the fluid; the flow combining end of the second flow divider/combiner is configured to receive the fluid outputted by at least one of the plate heat exchanger, the radiator and the heater, and the flow dividing end of the second flow divider/combiner is configured to output the fluid to a battery for thermal management of the battery; and the flow combining end of the first flow divider/combiner is configured to receive the fluid after thermal management of the battery; wherein the first flow divider/combiner and the second flow divider/combiner both are provided with normally closed drain valves, configured to conduct drainage and exhaust of a connecting pipeline connected to the first flow divider/combiner and the second flow divider/combiner.
 2. The liquid cooling unit according to claim 1, wherein the liquid cooling unit further comprises a compressor, a condenser and a cooling fan; the compressor, the condenser, the plate heat exchanger and the cooling fan are configured to cool a refrigerant; and a refrigerant pipeline connected to the compressor, the condenser and the plate heat exchanger are integrally welded.
 3. The liquid cooling unit according to claim 2, wherein the liquid cooling unit further comprises: a fluid chamber, the fluid chamber being a closed fluid chamber and being configured to provide a fluid to the second flow divider/combiner.
 4. The liquid cooling unit according to claim 3, wherein the liquid cooling unit further comprises: a water pump, the water pump being configured to output the fluid to the battery.
 5. The liquid cooling unit according to claim 4, wherein the water pump is a brushless direct current water pump.
 6. The liquid cooling unit according to claim 1, wherein there are a plurality of heaters.
 7. The liquid cooling unit according to claim 1, wherein the plate heat exchanger, the radiator and the heater are connected to the first flow divider/combiner and the second flow divider/combiner through rubber hoses.
 8. The liquid cooling unit according to claim 1, wherein the first flow divider/combiner and the second flow divider/combiner both are integrally formed by welding.
 9. The liquid cooling unit according to claim 1, wherein in a case where an environment temperature is lower than a first threshold, the cooling fan is started to cool the fluid inside the radiator.
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